

# Anushka Sheoran

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## EDUCATION

**University of Pennsylvania** | MSE in Computer Science | GPA: 3.9 Aug 2025- June 2027  
**University of California, San Diego** | B.Sc. Bioinformatics, Minor: Computer Science Sept. 2021 - June 2025  
**Relevant Coursework:** Agentic AI, Internet & Web Systems, Applied Machine Learning, Design & Analysis of Algorithms, Stats

## PROFESSIONAL EXPERIENCE

**Anthropic & University of Pennsylvania** Philadelphia, PA, USA  
*Claude Builder Club President & Claude Ambassador* Dec 2025 - Present

- Lead 1,000+ member AI development community as President and Anthropic Ambassador; drove 60% membership growth in 2 months across Wharton, SEAS, and professional schools. Maintained relationship with Anthropic team.
- Built and demoed agentic systems using Claude API, MCP, and multi-agent tool-use architectures to 100+ participants,

**Clariem Consulting** Mississauga, ON, CA  
*Product & Operations Analyst* May 2022 - Present

- Platform Engineering:** Shipped a LMS platform on LearnWorlds serving 50+ enterprise clients across 6 programs; integrated HubSpot CRM for 500+ contacts, automated personalized assessment via Zapier & Power Automate
- Agentic Product:** Led product lifecycle for [Launch Copilot](#), an LLM-powered system for drug launch teams, defined requirements with stakeholders, designed agentic workflow architecture, scoped MVP features, and shipped iteratively
- Data Product:** Defined requirements and delivered a Power BI analytics product for a Fortune 500 pharma client, building data pipelines unifying 10+ fragmented inputs into a single source of truth for launch tracking across functions

**FAIR Data Informatics (FDI) Lab** La Jolla, CA, USA  
*Data Curator/Analyst* Oct 2022 - Aug 2024

- Built clinical data harmonization pipelines across 35+ SCI datasets; designed ontology mapping workflows formally adopted by ODC-SCI consortium and fed to NIH TBI variables; published in Experimental Neurology

## PROJECTS

**Cloud-Based Web Search Engine** | *Java, Python, Apache Spark, AWS EC2, HTML* Sep 2025 - Jan 2026

- Architected distributed search engine on AWS EC2: custom multi-threaded web crawler, fault-tolerant key-value store with replication and failure recovery, and batch PageRank computation via Apache Spark
- Built inverted indexer with custom stemmer, TF-IDF scoring, and batch processing across 300k+ pages; implemented distributed PageRank with iterative convergence and batch writes to prevent JVM memory overflow
- Optimized Front-end:** Designed and built frontend search interface with TF-IDF + PageRank composite ranking, Levenshtein-distance spell checking, exact phrase match boosting, and HTML snippet generation with pagination

**Multi-Turn Medical AI Safety Evaluation** | *Python, OpenAI API, HuggingFace* Jan 2026 - Present

- Built MultiTurnPSB, a multi-agent evaluation pipeline that auto-generates 4-turn adversarial conversations, runs frontier models as defenders, and scores outputs via an LLM-as-a-judge; releasing frozen dataset on HuggingFace
- Engineered a live red-teaming agent pipeline configuring GPT-4o-mini as an autonomous attacker against GPT-4.1-mini and Claude Sonnet 4.5; surfaced a 19x unsafe-response divergence by Turn 4 invisible to single-turn evaluation
- Built an input-side classifier defense system that prepends category-aware safety tags to incoming queries; reduced Turn 4 unsafe outputs by 52.2pp despite classifier accuracy degrading to 48.5% under adversarial context

**Image2GPS: Vision-Language Geolocation System** | *Python, PyTorch, ResNet/ConvNeXt* Oct 2025 - Jan 2026

- Built end-to-end computer vision pipeline to predict GPS coordinates from campus images; curated 1000+ image dataset with EXIF metadata extraction, coordinate normalization, and train/val/test splits published to HuggingFace
- Iteratively improved architecture from ResNet-18 to ResNet-50 to ConvNeXt + k-NN hybrid with feature-space retrieval; applied geographically safe augmentations, validation-based checkpointing, AdamW with StepLR scheduling
- Reduced average localization error from 81.7m to 24.39m via GridSearch hyperparameter tuning and a hybrid ConvNeXt + k-NN inference mode that cut median external error from 42.84m to 30.91m

**Agentic AI Benchmarking Study: Claude 3.5 Sonnet on SWE-bench Verified** Jan 2026 - April 2026

- Reproduced Claude 3.5 Sonnet on SWE-bench Verified (500-task benchmark), achieving 20% resolve rate on 50 tasks; analyzed 27 failures using structured behavioral taxonomy, surfacing Tool errors (44%) and Verification failures (33%)
- Designed and evaluated 2 prompt-based interventions (TDD protocol, simplified verification); found that step budget and prompt design interact non-trivially, stricter constraints shifted failure distribution without improving resolve rate

## SKILLS

- Programming/ AI:** Python, Java, TypeScript, PyTorch, TensorFlow, scikit-learn, Pandas, Spark, SQL, LLMs, RAG
- Development:** React, REST APIs, FastAPI/Flask, Docker, CI/CD, AWS EC2, Git, Linux, Bash, Power BI, Excel